# Sentiora

A Permission-Based Digital Life Intelligence Platform

**Product Requirements Document**

Version 2.1 | Status: FINAL — Frozen for Technical Specification

Prepared for: Sentiora Product & Engineering Team

Date: July 24, 2026

# Document Control

| FIELD | VALUE |
| --- | --- |
| Document | Sentiora Product Requirements Document |
| Version | 2.1 |
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| Product | Sentiora |
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| Document type | Product Requirements Document (PRD) — precedes the Technical Specification |

**Product Requirements Frozen ≠ Stakeholder Sign-off Completed**

This document's product requirements are frozen as of v2.1 and are ready to serve as the source of truth for design and engineering. Formal stakeholder sign-off (Section 30) is a separate, still-pending step — signatures are intentionally blank until actual review occurs.

## Revision History

| VERSION | DATE | SUMMARY OF CHANGES |
| --- | --- | --- |
| 1.0 | Prior draft | Initial internal concept notes for Sentiora Memory and early product framing. |
| 2.0 | July 23, 2026 | Full company-grade PRD. Formalizes the four-module structure (Memory, Intelligence, Footprint, Shield), defines MVP v0.1 scope (Chrome-only capture, RAG search, no Footprint/Shield), and separates product requirements from technical implementation. |
| 2.1 | July 24, 2026 | **FINAL — Frozen for Technical Specification.** Resolved the MVP source-scope contradiction (FR-PRIV-001 now matches MVP Scope: webpage, PDF, YouTube, manual save only; AI conversations and GitHub moved cleanly to V1.x). Resolved the summaries priority contradiction (grounded summaries now consistently V1.x/P1; MVP Intelligence = Semantic Search + RAG Chat + Source Citations). Simplified the Meaningful Capture Engine requirement to remove a user-facing configurable-threshold claim from MVP. Clarified the sensitive-content exclusion policy in realistic, non-overclaiming terms, with detection mechanisms deferred to the Technical Specification. Added MVP internal delivery sequencing (Alpha/Beta/V1.x) without changing MVP scope. Categorized MVP success metrics and deferred numeric thresholds to Alpha/Beta baseline testing. Added a concise MVP value proposition statement. Converted Section 27 into a Decisions Register, resolving product-level open decisions and explicitly deferring technical ones. Added an MVP Exit/Freeze definition. Performed a full P0/P1/P2 consistency audit and a formatting/numbering/terminology pass. |

This document defines **what** is being built, **why** it matters, **who** it serves, and **what behavior is expected**. Architecture, schemas, APIs, and infrastructure decisions belong in the companion Technical Specification (see Section 28).

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# Executive Summary

Sentiora is a permission-based Digital Life Intelligence Platform that helps people remember, understand, organize, search, and protect their digital lives. It captures meaningful information only from sources a user explicitly approves, turns that information into searchable, structured personal memory, and layers reasoning, footprint awareness, and protection on top of it.

Modern digital life is fragmented across browsers, documents, videos, AI conversations, and dozens of connected services. Useful information — an article half-read last week, a research paper referenced in a chat, a concept explained in a YouTube video — becomes difficult to rediscover, connect, or act on. At the same time, most people have very little visibility into where their digital identity actually lives or what is quietly at risk.

Sentiora addresses this with four modules built on a single, strict principle: **permission before collection**.

> **MVP Value Proposition**
>
> Sentiora remembers the digital information you meaningfully engage with and lets you find and question it later — with your permission and with verifiable sources.

The long-term conceptual progression stays intact across all four modules: **Remember → Understand → Become Aware → Protect**, mapped to **Memory → Intelligence → Footprint → Shield**. The MVP delivers the first two steps of that progression.

| MODULE | ROLE | MVP STATUS |
|---|---|---|
| **Memory** | Remember what matters — consent-based capture of meaningful information from approved sources. | In MVP |
| **Intelligence** | Understand what you know — semantic search, RAG chat, and grounded, cited answers over Memory. | In MVP |
| **Footprint** | Understand where you exist online — maps accounts, services, and identity surface. | Next phase |
| **Shield** | Protect your digital identity — consumes Footprint + Memory signals to flag risk and recommend action. | Next phase |

The MVP (v0.1) is intentionally narrow: a Chrome-only browser extension covering consent-based capture from webpages, PDFs, and YouTube, manual save, a memory timeline, semantic search, and a RAG chat with citations. Footprint and Shield are explicitly out of scope for v0.1 and are staged for later phases once Memory and Intelligence have proven their core value.

Sentiora's differentiation is not "more data" — it is a **Meaningful Capture Engine** that filters for genuine engagement (dwell time, scroll depth, return visits) rather than logging everything indiscriminately, paired with a product principle that guides every decision in this document: "*Sentiora remembers what helps you, not secrets that could harm you.*"

# Problem Statement

### Formal Problem Statement

People generate valuable digital information every day, but it becomes fragmented across platforms, difficult to rediscover, difficult to understand collectively, and difficult to protect. Existing tools store activity or notes in isolation, leaving users without a single, trustworthy, permission-based layer that turns their own digital life into searchable knowledge — or gives them visibility into where that life is exposed.

## Specific Product Problems

| PROBLEM | DESCRIPTION |
|---|---|
| Information fragmentation | Meaningful content is scattered across browser tabs, PDFs, videos, notes, and AI chat threads with no shared memory layer. |
| Digital memory loss | Users forget where they saw something useful within days or weeks, even when it mattered to them at the time. |
| Difficulty rediscovering information | "I know I read this somewhere" is a common, unresolved moment with no efficient way to search past exposure. |
| Repeated searching | The same research or comparison (tools, papers, products) is redone from scratch because prior findings were never captured. |
| Information overload | Volume of daily content exceeds a person's capacity to manually organize or file it. |
| Lack of context between items | Related information encountered at different times and in different apps is never connected. |
| Difficulty extracting insight | Raw accumulated content (bookmarks, tabs, history) rarely turns into usable understanding on its own. |
| Lack of footprint visibility | Users don't know which accounts, services, and permissions make up their digital presence. |
| Unclear data exposure | It is hard to know which services actually hold personal information, and how current or risky that exposure is. |
| No unified intelligent interface | There is no single place to ask natural-language questions grounded in one's own digital knowledge. |

# Product Vision

Sentiora's long-term vision is to become the **Personal Intelligence Layer for a user's digital life** — a trusted, permission-based system that remembers what a person has seen and learned, helps them reason over it, and helps them understand and protect their presence across the digital world.

## Illustrative Future Queries

As Memory and Intelligence mature, a user should eventually be able to ask Sentiora natural-language questions such as:

- "Where did I see that article about vector databases?"
- "What was the research paper I read last month?"
- "Summarize everything I explored about RAG recently."
- "What topics have I been researching this month?"
- "Find the document where I saw this concept."
- "What do I already know about this subject?"
- "How are these pieces of information connected?"
- "Which digital services have I connected?"
- "What information has Sentiora stored about me?"
- "Delete everything collected from this source."

Every one of these answers must be grounded in the user's own authorized information — Sentiora surfaces and reasons over what the user chose to let it see; it does not claim knowledge it cannot verify.

# Product Goals & Non-Goals

## Goals

- Give users a single, trustworthy place to capture and rediscover meaningful digital information.
- Make personal knowledge searchable in natural language, with answers grounded in cited sources.
- Keep every act of collection explicit, visible, and reversible.
- Filter for meaningful engagement rather than logging everything a user does.
- Lay a foundation that can later extend into footprint awareness and protective recommendations without changing the trust model.

## Non-Goals (v0.1)

- Sentiora is **not** spyware, employee monitoring software, surveillance software, a keylogger, or a screen recorder.
- Sentiora does **not** indiscriminately track or store all browsing activity.
- Sentiora does **not**, in v0.1, map digital footprint or generate protective/security recommendations — that is Footprint and Shield, both next-phase.
- Sentiora does **not** attempt browser support beyond Chrome in v0.1.
- Sentiora does **not** capture passwords, payment details, banking information, health information, or content viewed in Incognito mode, under any configuration.

# Sentiora Platform Overview & Modules

Sentiora is the platform. It is composed of four modules, each with a distinct purpose, that together form the permission-based intelligence layer described in the vision. Memory and Intelligence make up the MVP. Footprint and Shield are approved, finalized directions for later phases.

### Sentiora Memory — "Remember what matters"

**Purpose:** Capture, structure, and organize meaningful information from sources the user has explicitly approved.

**Problem solved:** Digital memory loss and fragmentation — information seen once is otherwise lost.

**Core capabilities:** consent-based capture UI with granular per-source toggles; a Meaningful Capture Engine that filters for engagement (dwell time, scroll depth, revisits) instead of logging everything; a permanent, non-overridable sensitive-content exclusion policy covering Incognito mode, authentication/password-sensitive interfaces, payment and financial-sensitive interfaces, and private health-sensitive content where identifiable by supported detection rules; manual save of notes and links; memory timeline.

**Example interaction:** User reads a long article on vector databases and stays on the page for several minutes; Sentiora captures it as a structured memory with title, source, and key excerpt. A page glanced at for two seconds is not captured.

**Inputs:** approved browser pages, PDFs, YouTube videos, and manually added notes/links (MVP sources); AI conversations and GitHub content are approved V1.x sources, added once the MVP ingestion pipeline is proven — each on its own per-source toggle.

**Outputs:** structured, searchable memory records with metadata (source, timestamp, excerpt, topic tags).

**Dependencies:** none — Memory is the foundational layer other modules build on.

**Privacy implications:** highest-sensitivity module; governed entirely by explicit, granular, revocable consent and the never-capture list.

**MVP capability:** Yes — core of v0.1 **Future:** additional source connectors beyond the MVP set.

### Sentiora Intelligence — "Understand what you know"

**Purpose:** Reason over Sentiora Memory to answer questions, summarize, and surface connections.

**Problem solved:** Difficulty extracting insight and repeated searching for the same information.

**Core capabilities:** semantic search; natural-language Q&A via Retrieval-Augmented Generation (RAG); summarization; source citations back to the original memory.

**Example interaction:** User asks "What have I learned about vector databases recently?" — Sentiora retrieves relevant approved memories and generates a grounded answer with citations back to the original pages.

**Inputs:** the Memory index (embeddings + metadata); the user's natural-language query.

**Outputs:** grounded, cited answers; ranked search results; summaries.

**Dependencies:** depends entirely on Memory — Intelligence has nothing to reason over without it.

**Privacy implications:** operates only on already-approved Memory content; must never fabricate claims not supported by retrieved evidence.

**MVP capability:** Yes — semantic search + RAG chat with citations **Future:** topic clustering, proactive recommendations, cross-memory connection discovery.

### Sentiora Footprint — "Understand where you exist online"

Next phase — not in MVP

**Purpose:** Map the accounts, services, authentication methods, and identity surface a user has across the internet, separate from browsing activity or history.

**Problem solved:** Lack of visibility into which services hold personal information and how a user's digital identity is distributed.

**Core capabilities (planned):** account/service discovery through explicit connectors; permission and auth-method inventory; identity graph view.

**Realistic scope:** Footprint reflects what users explicitly connect or disclose — it does not claim to discover every account or piece of personal data anywhere on the internet.

**Dependencies:** can operate with its own connectors, but is enriched by Memory where relevant (e.g., a service mentioned in captured content).

**Privacy implications:** same explicit-consent model as Memory, scoped to account/service metadata rather than content.

### Sentiora Shield — "Protect your digital identity"

Next phase — not in MVP

**Purpose:** Consume signals from Footprint and Memory to flag risk and recommend protective action.

**Problem solved:** Users have no proactive help managing exposure or responding to risk in their digital footprint.

**Core capabilities (planned):** risk flagging (e.g., exposed or stale accounts, weak recovery paths); recommended actions (e.g., review permissions, revoke unused access).

**Key constraint:** Shield does not independently collect data — it is entirely downstream of Footprint and Memory.

**Dependencies:** depends on Footprint (primary) and Memory (secondary) — cannot function standalone.

**Privacy implications:** inherits the consent boundaries of its upstream modules; recommendations, not automated actions, unless the user explicitly authorizes an action.

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This four-module structure (Memory, Intelligence, Footprint, Shield) reflects the finalized product direction. Memory and Intelligence constitute the MVP; Footprint and Shield are approved for later phases and are described here at a product-intent level only — their detailed requirements will be developed once MVP learnings are in.

# Product Ecosystem Diagram

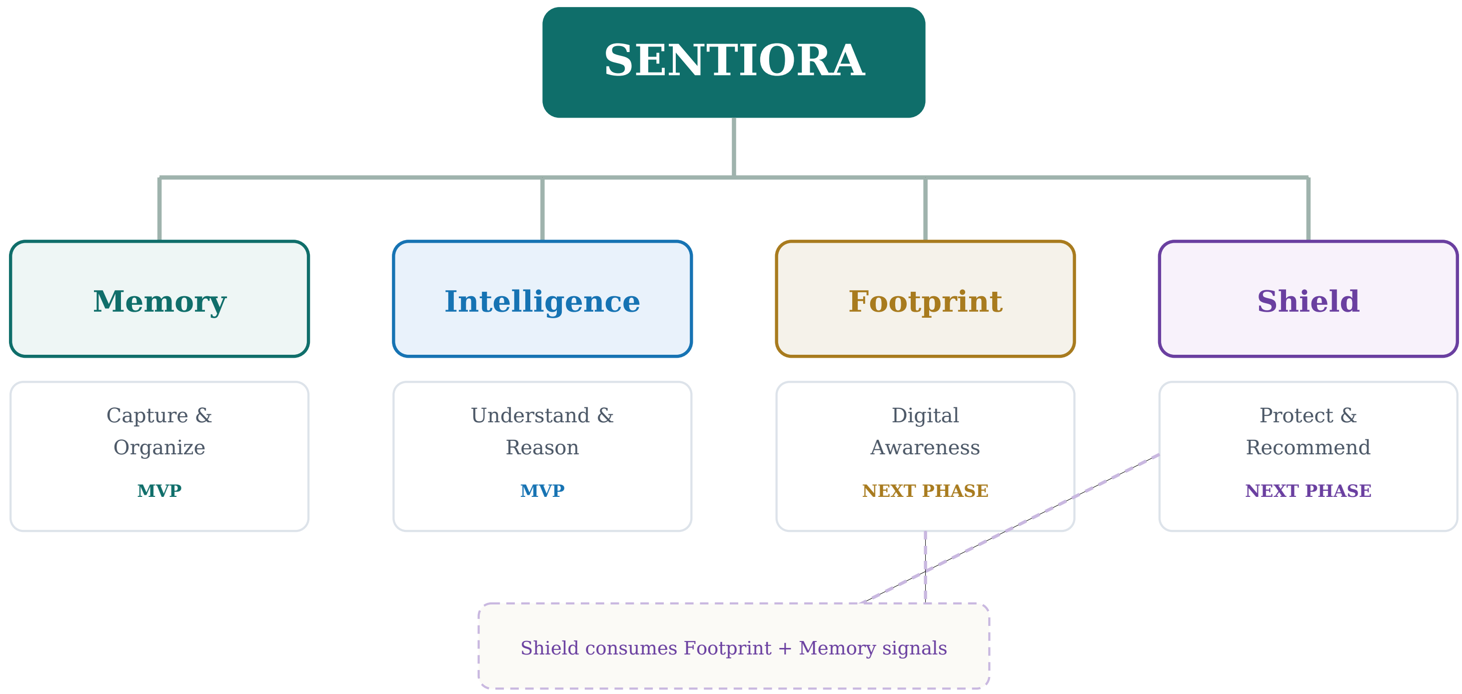


*Figure 1 — Sentiora Product Ecosystem. Memory and Intelligence are live in the MVP; Footprint and Shield are approved next-phase modules, with Shield depending on Footprint and Memory rather than collecting data independently.*

# Target Users & Personas

Sentiora's early-adopter segment is people who consume large amounts of research-style digital content and repeatedly struggle to rediscover it: students, researchers, developers, and knowledge workers. This is not "every internet user" — it is a segment with a frequent, painful, and recognizable version of the core problem.

### Primary Persona — The Research-Heavy Knowledge Worker

**Goals:** stay current in a fast-moving field; avoid re-researching the same ground twice.

**Problems:** reads dozens of articles, papers, and videos weekly across many tabs and tools; can't recall where a specific idea came from.

**Current behavior:** relies on scattered bookmarks, browser history, and memory; frequently re-searches.

**Sentiora use cases:** passive capture of meaningfully-read pages; RAG chat to answer "what have I read about X"; source citations for reuse in their own writing.

**Expected value:** less time re-finding, more time synthesizing.

### Primary Persona — The Student / Researcher

**Goals:** build durable understanding of a subject across a semester or research project.

**Problems:** content is spread across PDFs, papers, and videos; connections between sources are lost over time.

**Current behavior:** manual note-taking, folder systems, citation managers that don't capture browsing or video content.

**Sentiora use cases:** capture of PDFs and YouTube lectures; semantic search across a term's worth of material; grounded summaries with citations for review.

**Expected value:** a searchable second memory of everything they've actually engaged with.

### Secondary Persona — The Developer

**Goals:** track technical references, GitHub discussions, and documentation encountered while building.

**Problems:** loses track of which Stack Overflow answer or repo solved a past problem.

**Current behavior:** browser bookmarks, scattered notes, tab hoarding.

**Sentiora use cases:** capture of technical pages and GitHub content; quick recall via search when a similar problem resurfaces.

**Expected value:** faster problem-solving by reusing past research instead of repeating it.

Sentiora does not position itself as a mass-market tool for casual browsing at this stage. The early-adopter segment above is where the core value proposition — meaningful capture plus grounded recall — is felt most acutely and validated fastest.

# User Stories

| PRI. | AREA | STORY |
| --- | --- | --- |
| P0 | Onboarding | As a new user, I want a clear explanation of what Sentiora does and does not collect, so that I can decide whether to trust it before connecting anything. |
| P0 | Permissions | As a user, I want to approve capture on a per-source basis, so that I control exactly what Sentiora can see. |
| P0 | Memory capture | As a user, I want Sentiora to automatically capture pages I meaningfully engage with, so that I don't have to manually save everything. |
| P0 | Memory capture | As a user, I want to manually save a note or link at any time, so that important items are never missed even without passive capture. |
| P0 | Privacy controls | As a user, I want a permanent list of sensitive page types that are never captured, so that passwords, banking, and health information stay untouched regardless of settings. |
| P0 | Search | As a user, I want to search my memory using natural language, so that I can find something even if I don't recall the exact words. |
| P0 | AI questions | As a user, I want to ask Sentiora a question and get an answer grounded in my own memory, so that I can trust the response is based on what I actually saw. |
| P0 | Source citations | As a user, I want every AI-generated answer to cite the original source, so that I can verify and revisit it. |
| P0 | Memory management | As a user, I want to browse a timeline of everything Sentiora has captured, so that I stay aware of what is stored. |
| P0 | Data deletion | As a user, I want to delete an individual memory at any time, so that I retain full control over my stored information. |
| P1 | Connected sources | As a user, I want to see and manage which sources are currently connected, so that I can revoke access at any time. |
| P1 | Summaries | As a user, I want a summary of everything I've explored on a topic recently, so that I can quickly reorient myself. |
| P1 | Dashboard | As a user, I want a home dashboard showing recent memories and quick search, so that I can start each session productively. |
| P1 | Data export | As a user, I want to export my memory data, so that I retain ownership of my information outside Sentiora. |
| P2 | Digital footprint | As a user, I want to see which accounts and services make up my digital footprint, so that I understand my broader digital presence. |
| P2 | Protection | As a user, I want Sentiora to flag risky or stale accounts, so that I can take protective action. |

# End-to-End User Journey

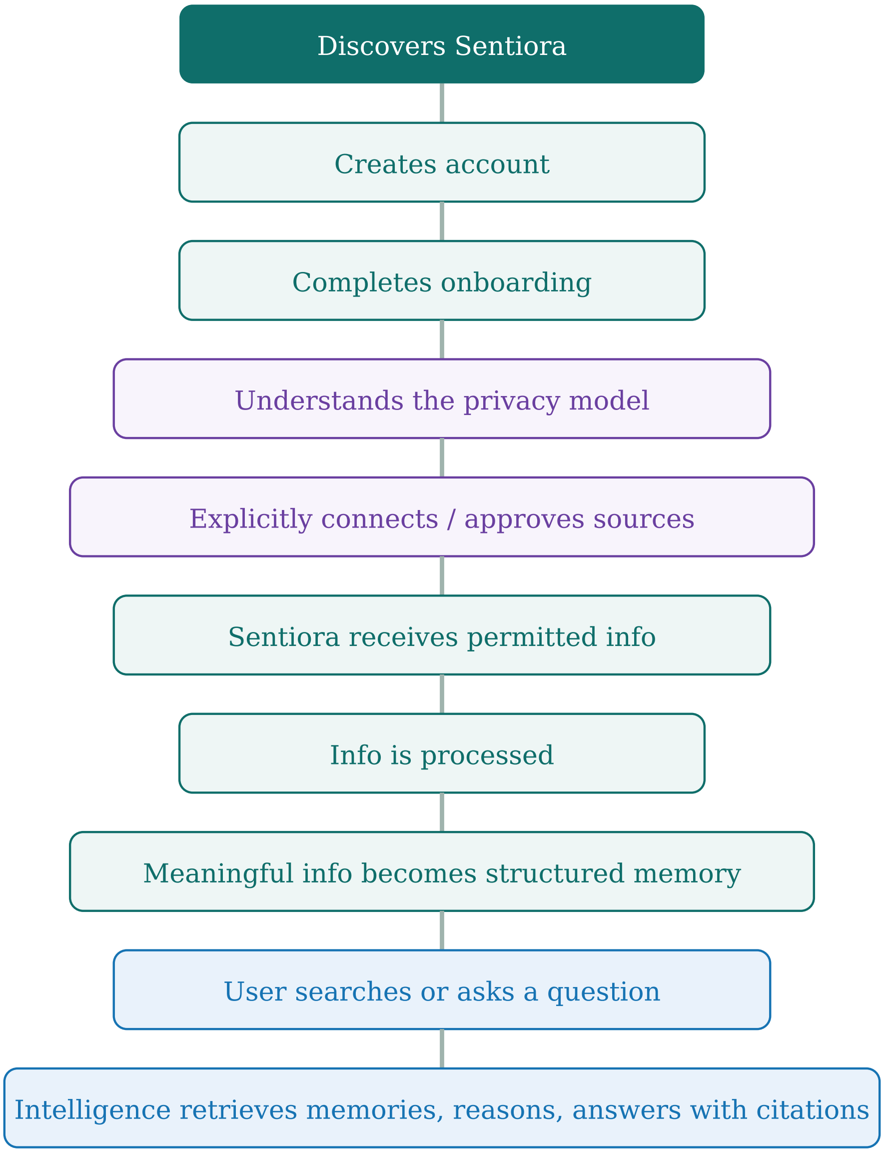


*Figure 2 — End-to-End User Journey. Purple steps are privacy/consent moments; green steps are Memory; blue steps are Intelligence. After an answer is returned, the user can always inspect, manage, or delete any memory referenced.*

# Information Lifecycle Flowchart

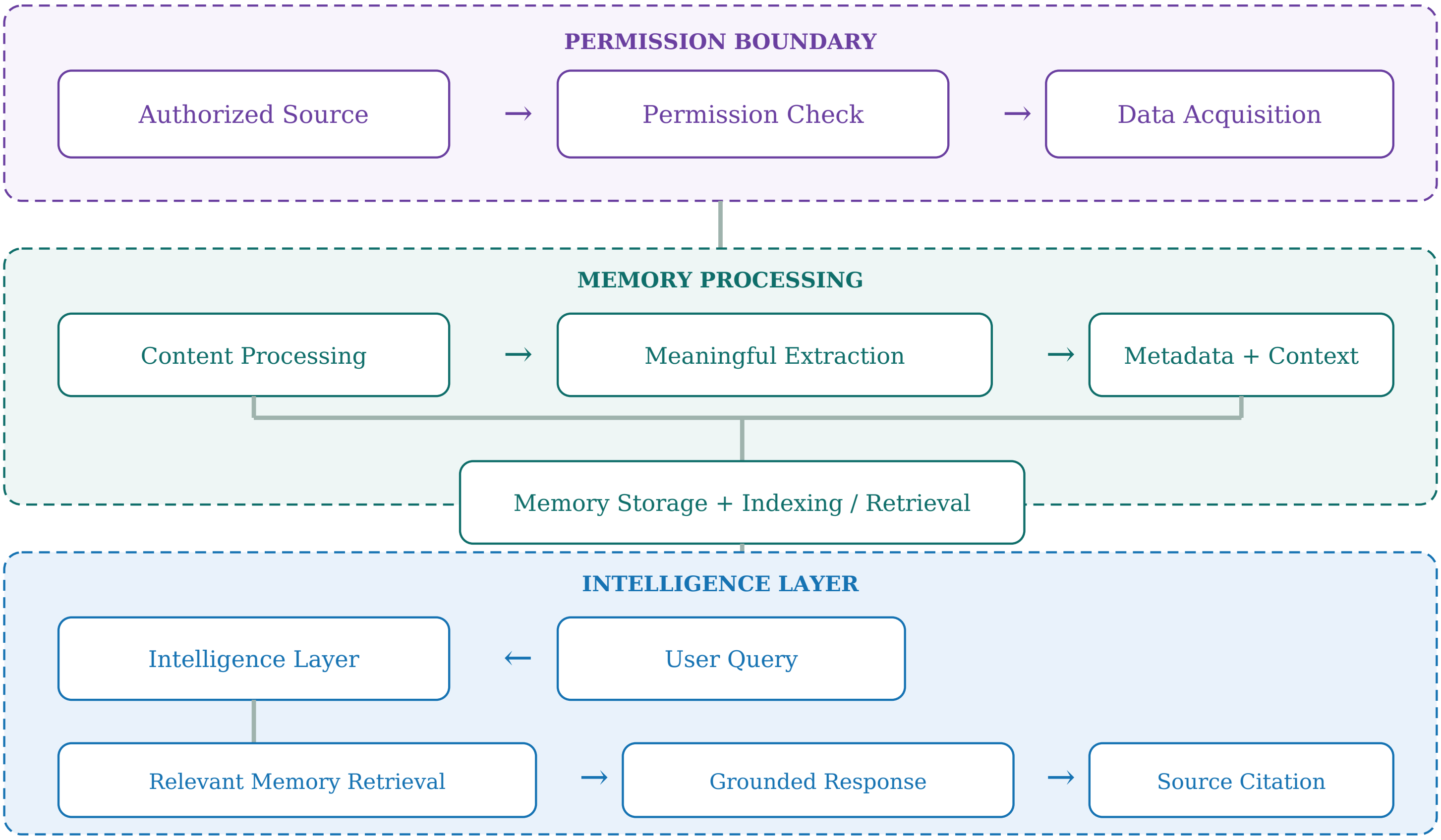


*Figure 3 — Information Lifecycle. Dashed boundaries mark the permission boundary (purple), Memory processing (green), and the Intelligence layer (blue) — nothing crosses from acquisition into storage without first clearing the permission check.*

# Permission & Privacy Flow

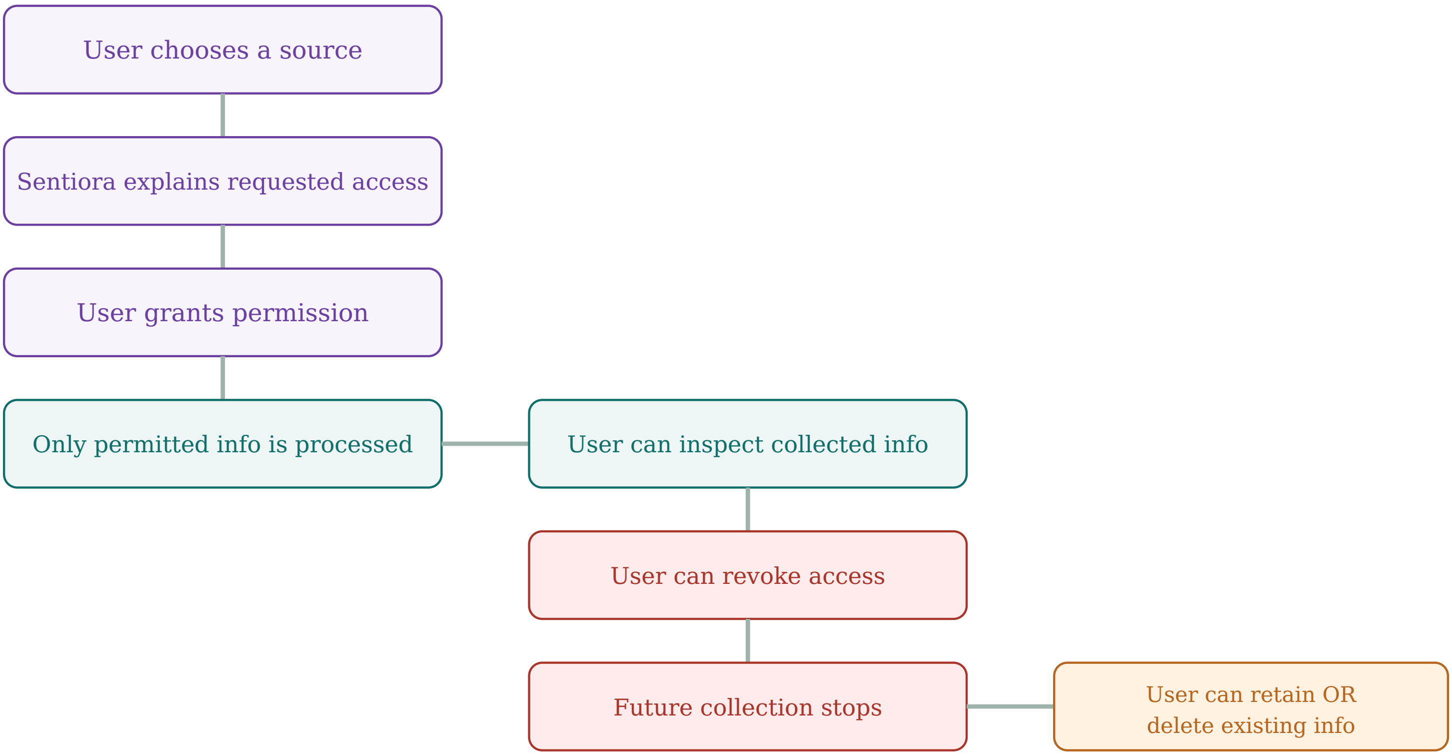


*Figure 4 — Permission & Privacy Flow. Note the explicit branch at the end:* ***revoking access stops future collection but does not automatically delete previously stored memories*** *— deletion is always a separate, user-initiated choice.*

**Key Distinction**

**Revoke access ≠ automatically delete previously stored information.** Revoking a source's permission stops all future capture from that source immediately. Existing memories already captured under prior permission remain unless the user separately chooses to delete them, individually or in bulk, from the Privacy Center.

# Functional Requirements

Requirements are grouped by area and identified with a stable ID for traceability into the Technical Specification and future test plans.

## Authentication & Onboarding

| ID | REQUIREMENT | DESCRIPTION | PRI. |
|---|---|---|---|
| FR-AUTH-001 | Account creation | User can create an account via email or supported SSO. *Acceptance:* account created and verified before any source connection is offered. | P0 |
| FR-UI-001 | Consent-first onboarding | Onboarding must explain the permission model in plain language before any capture is enabled. *Acceptance:* capture cannot be toggled on until the privacy explanation screen has been viewed. | P0 |

## Source Connection & Permission Management

| ID | REQUIREMENT | DESCRIPTION | PRI. |
|---|---|---|---|
| FR-PRIV-001 | Per-source toggles | User can enable/disable capture independently for each MVP-supported source: webpages, PDFs, YouTube, and manual notes/links. *Acceptance:* disabling one source has no effect on others. AI conversation capture and GitHub capture are V1.x sources (see FR-PRIV-001a) and are out of scope for this requirement's MVP acceptance test. | P0 |
| FR-PRIV-001a | Per-source toggles — V1.x sources | User can enable/disable capture independently for AI conversations and GitHub once these sources are introduced in V1.x. *Acceptance:* deferred to V1.x; not required for MVP v0.1. | P1 |
| FR-PRIV-002 | Sensitive-content exclusion policy | System enforces a permanent, non-overridable exclusion policy at capture time: Incognito mode is never captured; authentication/password-sensitive interfaces are never captured; payment and financial-sensitive interfaces are never captured; private health-sensitive interfaces/content are excluded where identifiable by supported detection rules. *Acceptance:* no memory is created from Incognito, authentication, or payment/financial interfaces, verified by test pages in each category; health-sensitive exclusion is verified against the supported detection rules defined in the Technical Specification. Exact detection mechanisms, domain rules, classifiers, and false-positive handling are defined in the Technical Specification, not this PRD. | P0 |
| FR-PRIV-003 | Revoke access | User can revoke a connected source at any time. *Acceptance:* capture from that source stops within one session; previously stored memories remain unless separately deleted. | P0 |

## Data Ingestion & Memory Creation

| ID | REQUIREMENT | DESCRIPTION | PRI. |
|---|---|---|---|
| FR-MEM-001 | Meaningful Capture Engine | System evaluates engagement signals (such as dwell time, scroll depth, and revisits) against product-defined internal thresholds before creating a memory from a webpage. *Acceptance:* pages/items below the meaningful-engagement threshold do not automatically become memories; Manual Save remains available regardless of the automatic capture decision. The exact scoring formula and threshold values are a Technical Specification concern, not a user-facing setting in MVP. | P0 |
| FR-MEM-002 | PDF capture | User can capture a PDF opened in-browser as a memory with extracted text. *Acceptance:* captured PDF memory includes title, source path, and extractable text for indexing. | P0 |
| FR-MEM-003 | YouTube capture | System captures a YouTube video as a memory when engagement criteria are met, including title and available transcript. *Acceptance:* memory includes video metadata and transcript text where available. | P0 |
| FR-MEM-004 | Manual save | User can manually save a note or link at any time regardless of passive capture settings. *Acceptance:* manual save always succeeds and is timestamped. | P0 |

## Memory Browsing & Search

| ID | REQUIREMENT | DESCRIPTION | PRI. |
|---|---|---|---|
| FR-MEM-005 | Memory timeline | User can browse captured memories chronologically. *Acceptance:* timeline loads, is filterable by source type and date. | P0 |
| FR-SEARCH-001 | Semantic search | User can search memory using natural language, not just exact keyword match. *Acceptance:* a semantically related query returns relevant memories even without exact keyword overlap. | P0 |

## AI Q&A / RAG / Citations

| ID | REQUIREMENT | DESCRIPTION | PRI. |
|---|---|---|---|

| | | | |
|---|---|---|---|
| FR-INT-001 | RAG chat | User can ask natural-language questions and receive answers grounded in retrieved memory. *Acceptance:* every substantive claim in a response is traceable to at least one retrieved memory. | P0 |
| FR-INT-002 | Source citations | Every RAG response displays citations linking back to originating memories. *Acceptance:* clicking a citation opens the source memory. | P0 |
| FR-INT-003 | Grounded summaries | User can request a summary of memories on a topic. *Acceptance:* summary cites the memories it draws from and does not include unsupported claims. | P1 |

## Footprint Dashboard / Privacy Center

| ID | REQUIREMENT | DESCRIPTION | PRI. |
|---|---|---|---|
| FR-FOOT-001 | Footprint dashboard | Next-phase: user can view mapped accounts and services. *Acceptance criteria to be defined in the Footprint phase spec.* | Next phase |
| FR-PRIV-004 | Privacy Center | User can view all connected sources, all stored memories, and initiate deletion or export from one place. *Acceptance:* every stored memory is visible and actionable from this screen. | P0 |

## Data Deletion, Export & Account

| ID | REQUIREMENT | DESCRIPTION | PRI. |
|---|---|---|---|
| FR-PRIV-005 | Delete individual memory | User can delete a single memory permanently. *Acceptance:* deleted memory is removed from storage and search index within the session. | P0 |
| FR-PRIV-006 | Data export | User can export their memory data in a portable format. *Acceptance:* export includes memory content and metadata for all sources. | P1 |
| FR-PRIV-007 | Account deletion | User can delete their account, which deletes all associated memories. *Acceptance:* confirmed account deletion removes all user data within the stated retention window. | P1 |

## Error Handling

| ID | REQUIREMENT | DESCRIPTION | PRI. |
|---|---|---|---|
| FR-UI-002 | Ingestion failure feedback | User is notified when a capture or ingestion attempt fails. *Acceptance:* failed capture surfaces a visible, non-blocking notification with a retry option. | P1 |

# Non-Functional Requirements

| CATEGORY | REQUIREMENT (TARGET, NOT GUARANTEE) |
|---|---|
| Security | All data in transit is encrypted; access to stored memory requires authenticated sessions. |
| Privacy | No memory is created outside explicitly approved sources; the never-capture list is enforced at the point of capture, not after the fact. |
| Performance | Search and RAG responses should return within a few seconds under typical MVP load; this is a target for early testing, not a contractual SLA. |
| Reliability | Capture failures should degrade gracefully (skip and notify) rather than blocking browsing. |
| Scalability | Storage and indexing should scale per-user without requiring architectural rework before Footprint/Shield phases. |
| Accessibility | Core screens (timeline, search, chat, privacy center) should meet baseline accessibility practices (keyboard navigation, contrast, screen-reader labels). |
| Explainability | Every AI-generated answer must be traceable to cited source memories (see Section 16). |
| Data portability | Users can export their data in a structured, human-readable format. |
| Observability | Capture, ingestion, and deletion events should be logged for debugging and for user-facing activity transparency. |
| Maintainability | Module boundaries (Memory / Intelligence / Footprint / Shield) should remain independently deployable as the product grows. |

# Privacy by Design

Privacy is not a feature of Sentiora — it is the constraint every other feature is designed within. The following principles apply to every module, present and future.

| PRINCIPLE | WHAT IT MEANS IN SENTIORA |
|---|---|
| **Explicit Consent** | No source is accessed without the user's clear, affirmative authorization — no pre-checked boxes, no buried terms. |
| **Data Minimization** | Only information required for the selected functionality is collected; the Meaningful Capture Engine exists specifically to avoid collecting everything by default. |
| **Transparency** | Users can always see what was collected, from where, and when. |
| **User Control** | Users can inspect, disconnect, export, and delete their information at any time. |
| **Source-Level Permissions** | Permissions are managed independently per source wherever technically possible, not as a single all-or-nothing switch. |
| **Deletion** | Individual memories, source-specific data, or the full account can all be deleted. |
| **Security** | Sensitive information is protected in transit and at rest where applicable. |
| **No Hidden Surveillance** | Sentiora never monitors secretly; capture is always visible and attributable to a permission the user granted. |


*Figure 5 — Privacy Control Flow. Every stage feeds forward into user control: nothing is collected, stored, or reasoned over without a visible, revocable, and deletable trail back to user consent.*

# Search & Intelligence Experience

| MODE | EXPECTED BEHAVIOR |
|---|---|
| Keyword search | Returns memories containing matching terms, ranked by relevance and recency. |
| Semantic search | Returns conceptually related memories even without exact keyword overlap, using embedding similarity. |
| Natural-language questions | Routed to the RAG chat; retrieves relevant memories, then generates a grounded answer. |
| Memory retrieval | Retrieval is scoped strictly to the asking user's own approved memories. |
| Summaries | Generated only from retrieved memories, with citations attached. |
| Source citations | Attached to every generated claim, linking to the exact source memory. |

## Behavior in Boundary Conditions

| CONDITION | EXPECTED BEHAVIOR |
|---|---|
| Relevant information exists | Answer is generated with citations to the supporting memories. |
| Information is insufficient | Sentiora states that it doesn't have enough information rather than guessing. |
| Multiple conflicting memories | Both are surfaced with their sources rather than silently picking one. |
| Source is unavailable | Cached memory content is used; the response notes the original source may no longer be reachable. |
| Model is uncertain | Uncertainty is communicated explicitly rather than presented as confident fact. |

**Guiding rule:** Sentiora never presents an unsupported AI-generated claim as a fact drawn from the user's memory. If evidence isn't there, the product says so.

# Trust & Explainability

Trust is treated as a product requirement, not an afterthought. Every answer Sentiora gives should be checkable.

- **Source citations** — every generated claim links to the memory it came from.
- **Memory references** — citations include source name and capture date.
- **Relevant excerpt** — the specific passage used is shown, not just a link.
- **Confidence/uncertainty communication** — the system states when evidence is thin or absent.
- **"Why am I seeing this?" explanations** — search and recommendation results can show why a memory matched.

# Product Dashboard / Information Architecture

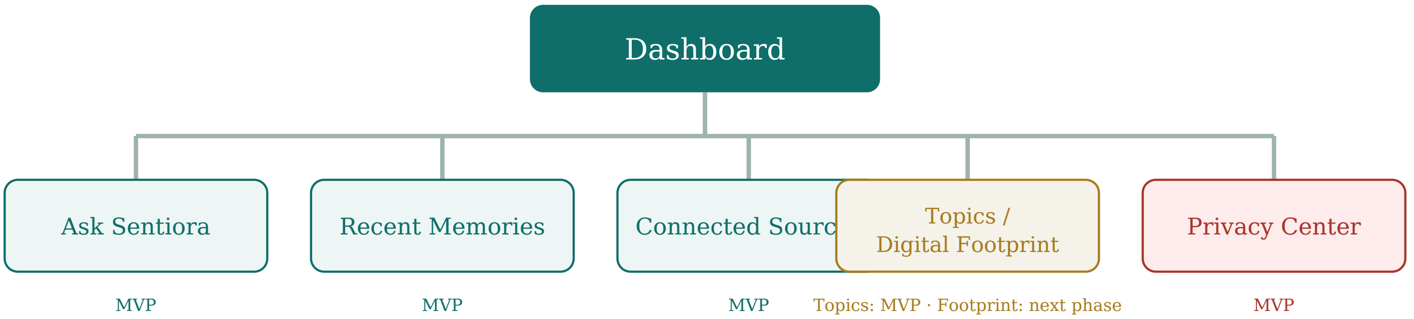


*Figure 6 — Dashboard Information Architecture. Ask Sentiora, Recent Memories, Connected Sources, Topics, and the Privacy Center are MVP; the Footprint view is staged for the next phase.*

Detailed UI mockups are intentionally out of scope for this PRD; this diagram establishes the conceptual structure the design team will build from.

# MVP Definition & Scope

The MVP (v0.1) is scoped for a small, early-stage team to ship and validate quickly: a Chrome-only extension proving the core loop of consent-based capture → structured memory → grounded search and chat.

## MVP / V1 — In Scope

| FEATURE | PRIORITY | COMPLEXITY | USER VALUE / REASON FOR INCLUSION |
|---|---|---|---|
| Chrome extension shell | P0 | Medium | Only viable single-browser surface for a four-person team to ship well. |
| Consent & capture controls | P0 | Medium | Core trust mechanism; without it nothing else is credible. |
| Never-capture list | P0 | Low | Non-negotiable safety guarantee. |
| Webpage capture (Meaningful Capture Engine) | P0 | High | Primary source of everyday value. |
| PDF capture | P0 | Medium | High-value for the research/student persona. |
| YouTube capture | P0 | Medium | Common research format not covered by note tools. |
| Manual save | P0 | Low | Reliable fallback independent of passive capture. |
| Memory timeline | P0 | Low | Baseline transparency and browsing. |
| Semantic search | P0 | High | Core differentiator over plain bookmarking. |
| RAG chat with citations | P0 | High | The headline "understand what you know" experience. |

## V1.x — Important Improvements

| FEATURE | PRIORITY | REASON |
|---|---|---|
| Grounded summaries with citations | P1 | Natural extension of RAG chat; valuable, but not required for the MVP's essential loop of Semantic Search + RAG Chat + Source Citations (FR-INT-003). |
| Data export | P1 | Reinforces user ownership once core loop is validated. |
| Account deletion flow | P1 | Completes the privacy control set. |
| AI conversation capture | P1 | Extends source coverage once ingestion pipeline is proven. |
| GitHub capture | P1 | Extends source coverage once ingestion pipeline is proven; sequencing relative to AI conversation capture to be decided from MVP user feedback. |
| Additional browser support | P1 | Expands addressable users past Chrome-only. |

## V2+ — Advanced Capabilities

| FEATURE | PRIORITY | REASON |
|---|---|---|
| Sentiora Footprint | P2 | Requires new connector architecture beyond MVP scope. |
| Sentiora Shield | P2 | Depends on Footprint being live first. |
| Topic clustering & proactive recommendations | P2 | Needs a mature memory corpus to be useful. |

### Explicitly Out of Scope for MVP

Footprint mapping, Shield risk flagging, browsers other than Chrome, mobile apps, automated account-level protective actions, and any form of always-on activity logging.

## Recommended Internal Delivery Sequence

The MVP is a meaningful build for a four-person team. The features above remain the full MVP scope; the sequence below is an internal delivery aid, not a separate product or a sprint plan.

| STAGE | INCLUDES |
|---|---|
| **MVP Alpha** | Chrome extension shell; consent-first onboarding; source permissions; never-capture enforcement; webpage capture; manual save; memory timeline; semantic search; RAG chat; source citations; individual memory deletion. |
| **MVP Beta** | PDF capture; YouTube capture; Meaningful Capture Engine refinement; Privacy Center refinement; ingestion/error-handling refinement. |
| **V1.x** | Grounded summaries; data export; account deletion flow; AI conversation capture; GitHub capture; additional browser support. |

Alpha and Beta are sequencing labels for the same MVP v0.1 scope — not separate releases with separate requirements.

## MVP Exit / Freeze Definition

MVP v0.1 is product-complete when a user can successfully complete the full core loop:

> Access Sentiora → understand consent → approve supported source capture → meaningfully engage with content → Sentiora creates a structured memory → user sees the memory → user searches naturally → user asks a question → Sentiora retrieves relevant memory → Sentiora returns a grounded answer → answer contains source citation → user can inspect/delete the memory

...and, at the same time:

- Never-capture rules are enforced.
- Revocation stops future capture.
- Deleted memories stop appearing in future retrieval.
- Insufficient evidence produces an honest "not enough information" response rather than a fabricated answer.
- Failed ingestion does not silently create corrupted memories.

This defines MVP completion at product acceptance level. Test plans, QA criteria, and engineering-level exit checks belong in the Technical Specification and launch readiness plan.

# MVP Success Criteria

Metrics are grouped by what they validate. No numeric go/no-go thresholds are set in this document.

## Acquisition / Activation

| METRIC | WHAT IT VALIDATES |
|---|---|
| Onboarding completion rate | Whether the consent-first flow is clear enough to get users to a usable state. |
| % of users connecting at least one source | Whether the value proposition is compelling enough to grant real access. |

## Memory Quality

| METRIC | WHAT IT VALIDATES |
|---|---|
| Number of meaningful memories generated per active user | Whether the Meaningful Capture Engine is producing useful, not noisy, output. |
| Capture usefulness / noise (user-reported or inferred) | Whether captured memories are ones the user actually wanted kept. |

## Retrieval Quality

| METRIC | WHAT IT VALIDATES |
|---|---|
| Search success rate (query → memory opened) | Whether semantic search actually surfaces the right thing. |
| Query answer usefulness (user rating or follow-up rate) | Whether RAG answers are actually helpful, not just plausible. |

## Trust

| METRIC | WHAT IT VALIDATES |
|---|---|
| Citation/source opening rate | Whether users trust and verify AI answers, a proxy for explainability working. |
| Successful deletion/revocation completion rate | Whether privacy controls work reliably when used. |

## Retention

| METRIC | WHAT IT VALIDATES |
|---|---|
| Weekly returning users | Whether the product earns a place in a person's regular routine. |
| Memory retrieval usage (searches/chats per active user) | Whether Intelligence, not just Memory, is delivering value. |

These are chosen to avoid vanity metrics (e.g., raw page views) in favor of signals tied directly to the core value loop: capture → retrieve → trust → return.

**Quantitative Thresholds**

Quantitative go/no-go thresholds will be established from Alpha/Beta baseline data before public MVP launch, and recorded in the launch readiness / test plan — rather than invented in this PRD.

# Edge Cases

| CASE | EXPECTED BEHAVIOR |
| --- | --- |
| Permission revoked mid-session | Capture from that source stops immediately; existing memories remain untouched. |
| Source unavailable at query time | Cached memory content is used; response notes the source may be unreachable. |
| Duplicate information captured twice | System deduplicates or links duplicates rather than showing repeated near-identical memories. |
| Original source later deleted online | Captured memory persists (it was already stored); response indicates the live source may no longer exist. |
| Empty memory (no sources connected yet) | Search and chat clearly explain there is nothing to search yet, with a prompt to connect a source. |
| Conflicting information across memories | Both are surfaced with sources; system does not silently resolve the conflict. |
| Unsupported file type | Capture fails gracefully with a clear, non-blocking notification. |
| Very large document | Content is chunked for processing; user is not blocked from browsing while it processes. |
| Failed ingestion | User is notified with a retry option (FR-UI-002); partial memory is not silently stored. |
| User deletes a memory referenced by a past chat answer | Past answer text remains as history, but re-asking the same question no longer retrieves the deleted memory. |
| AI cannot find enough evidence | System states it doesn't have enough information rather than fabricating an answer. |
| Source connection expires (auth token, etc.) | User is notified and prompted to reconnect; no silent failure. |
| Account deletion requested | All associated memories are deleted within the stated retention window; user receives confirmation. |

# Risks & Mitigation

| RISK | IMPACT | LIKELIHOOD | MITIGATION |
|---|---|---|---|
| Privacy risk | Loss of user trust; reputational and potential legal exposure. | Medium | Explicit consent model, never-capture list enforced at point of capture, transparent Privacy Center. |
| Security risk | Unauthorized access to stored memory. | Medium | Encryption in transit, authenticated access, minimal retained sensitive data. |
| Hallucination risk | AI states something not actually supported by the user's memory. | Medium-High | RAG grounding, mandatory citations, explicit "insufficient evidence" behavior. |
| Incorrect retrieval | Wrong or irrelevant memory surfaced, undermining trust. | Medium | Relevance tuning, visible source excerpts so users can self-verify. |
| Over-collection | Capturing more than intended erodes the "not surveillance" positioning. | Medium | Meaningful Capture Engine thresholds, per-source toggles, never-capture list. |
| User trust | Users abandon the product if consent model feels unclear or violated. | Medium | Consent-first onboarding, always-visible activity log, easy revocation and deletion. |
| Integration dependency | Reliance on third-party page/PDF/YouTube access patterns that may change. | Medium | Modular ingestion pipeline isolated from capture triggers; addressed in Tech Spec. |
| Scalability | Memory/index growth per user over time. | Low (at MVP scale) | Design storage/indexing to scale without architectural rework (see NFR). |
| AI cost | RAG and embedding calls scale with usage and could pressure unit economics. | Medium | Chunking and caching strategy; addressed in Tech Spec. |
| Data storage cost | Growing memory corpus increases storage spend. | Low-Medium | Meaningful Capture Engine limits stored volume to genuinely engaged content. |
| Source availability | Captured source pages may go offline or change. | Medium | Store sufficient captured content/metadata to remain useful even if the live source disappears. |
| Legal / compliance considerations | Data protection obligations vary by jurisdiction and are not yet formally assessed. | Unknown | Flagged as an open item for legal review before broader launch; no compliance claim is made in this document. |

# Competitive Positioning

Sentiora sits at the intersection of several existing categories without being fully replaced by any one of them.

| CATEGORY | HOW SENTIORA DIFFERS |
|---|---|
| Note-taking systems | Notes require manual entry; Sentiora also passively captures meaningfully-engaged content without manual effort. |
| Personal knowledge management (PKM) tools | PKM tools organize what a user manually inputs; Sentiora's Memory layer starts from what the user actually browsed and read. |
| General AI assistants | General assistants reason over public knowledge or a single session; Sentiora grounds answers in the user's own authorized history. |
| Browser history / search tools | Raw history is an unfiltered log; Sentiora filters for meaning and turns it into structured, searchable, cited knowledge. |
| Digital footprint / privacy tools | These typically focus only on exposure scanning; Sentiora pairs footprint awareness (future) with an actual memory and reasoning layer. |
| Personal AI memory systems | Closest category; Sentiora differentiates on explicit per-source consent, a never-capture list, and the planned Footprint/Shield extension. |

**Core differentiation:** *Memory + Intelligence + Digital Awareness + User Control* — no invented statistics, no unverifiable competitor claims; this section will be revisited with real market data as it becomes available.

# Product Principles

1. Permission before collection.
2. The user owns and controls their memory.
3. Useful information over indiscriminate data.
4. Evidence before AI confidence.
5. Privacy by design.
6. Explainable intelligence.
7. Easy deletion and revocation.
8. Intelligence should improve as useful memory grows.
9. Minimize unnecessary collection.
10. Never sacrifice user trust for additional data.

**Anchor Principle**

"Sentiora remembers what helps you, not secrets that could harm you."

# Product Roadmap

**All phase timing below is proposed, not guaranteed.** Sprint-level assignments belong in the development plan, not this PRD.

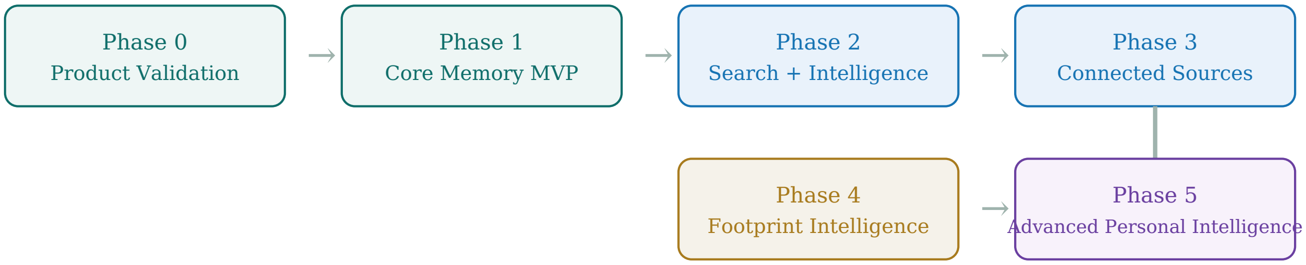


*Figure 7 — Proposed roadmap. Green = MVP phases, blue = Intelligence maturation, tan/purple = Footprint and Shield, staged after core validation.*

# Product Dependency Map

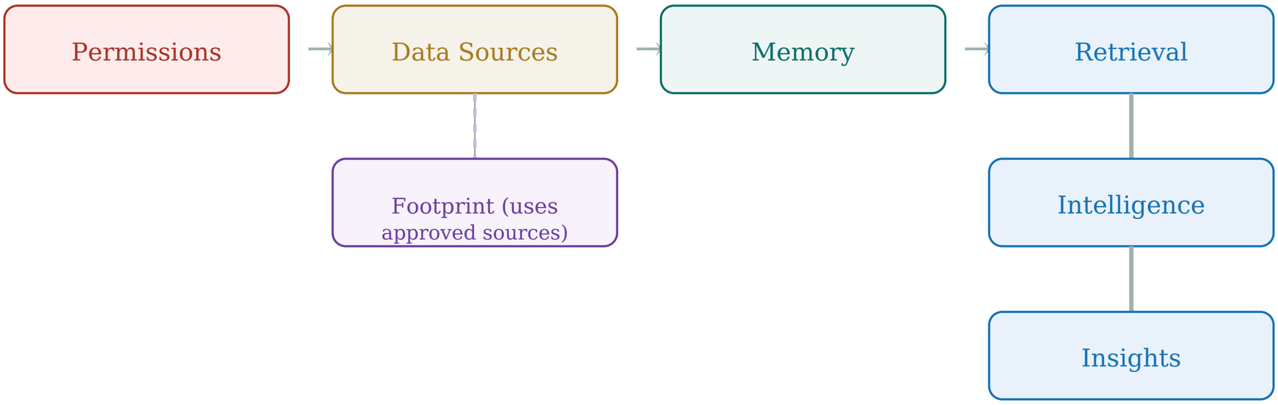


*Figure 8 — Product Dependency Map. Memory and Intelligence form the linear MVP dependency chain. Footprint can operate through its own connectors but is enriched by the same permission and source layer; Shield (not shown) depends on both Footprint and Memory and cannot run independently.*

# Data Control Model

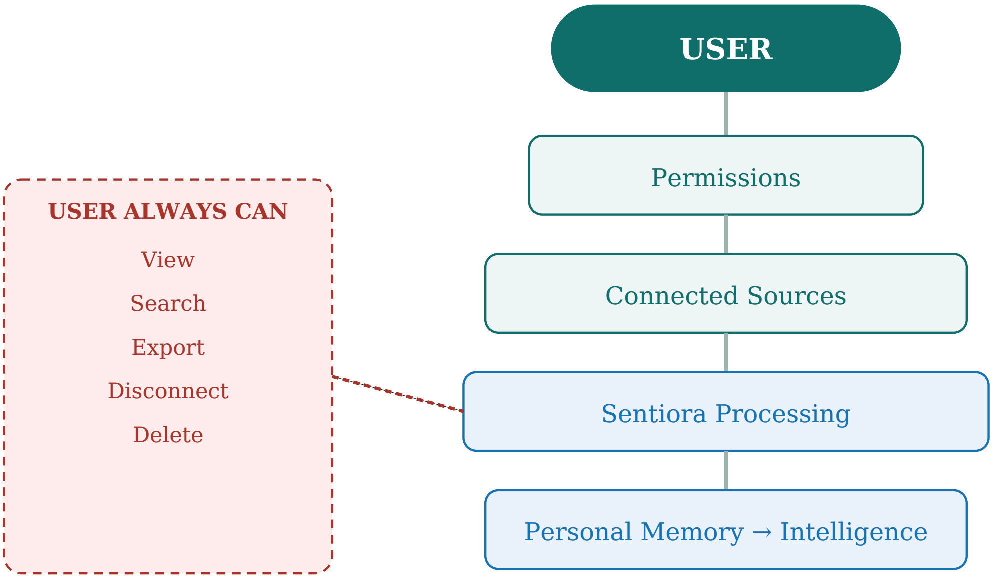


*Figure 9 — Data Control Model. The user sits above every stage of processing, with view, search, export, disconnect, and delete controls available against the pipeline at all times — reinforcing that the user remains the controlling entity, not a passive data source.*

# Open Product Decisions — Decisions Register

Freezing this PRD means every prior open question is now either a settled product decision, or explicitly deferred to the Technical Specification. Nothing here is left ambiguous.

## A. Product Decisions — Resolved at Freeze

**Data source sequencing:** MVP order is Webpage → PDF → YouTube → Manual Save. V1.x adds AI Conversations and GitHub; no priority is forced between the two — sequencing between them will be decided from MVP user feedback.

**Default retention:** Memories remain available until deleted by the user, or governed by an explicitly communicated future retention policy. No automatic deletion period is defined at product level.

**User control / deletion scope:** Users must be able to delete individual memories (MVP). Source-level and bulk deletion are future improvements. Account deletion ultimately removes associated user data according to the documented deletion/retention process.

**Sensitive memory classification:** The fixed, non-overridable sensitive-content exclusion policy (Section 14, FR-PRIV-002) is mandatory and unconditional. User-adjustable sensitivity controls beyond it are future exploration and are not required for MVP.

**Footprint MVP scope:** Intentionally deferred until the Footprint phase begins; not defined in this PRD.

**Additional consent tiers:** Any future capability that performs an action beyond passive analysis or recommendation (e.g., a Shield action) must require its own explicit user authorization, separate from the original source permission.

**Storage vs. reference (product-level constraint only):** Sentiora stores only the minimum content necessary to provide reliable memory retrieval, citations, and user control. Exact storage representation and architecture are defined in the Technical Specification.

## B. Technical Decisions — Deferred to Technical Specification

The following remain intentionally undecided here and are the Technical Specification's responsibility, not this PRD's:

- Exact local vs. cloud architecture
- Database / storage technologies and vector database selection
- Exact raw-content representation
- Embedding model and LLM/model provider
- Chunking strategy
- Encryption implementation
- Caching strategy
- Ingestion implementation
- Exact sensitive-content detection mechanism (domain rules, classifiers, false-positive handling)
- Infrastructure / deployment topology
- API architecture

# PRD → Technical Specification Handoff

This PRD defines product intent and behavior. The following must be answered by the companion Technical Specification, not here:

- System architecture (extension, backend services, processing pipeline).
- Database design and memory schema.
- Embedding strategy and vector storage choice.
- RAG architecture and retrieval pipeline.
- Authentication implementation.
- Encryption approach (in transit and at rest).
- Data ingestion architecture per source type.
- Source connector architecture (current and future sources).
- Permission enforcement mechanism at the technical layer.
- Deletion propagation across storage, index, and cache.
- API design between extension, backend, and AI layer.
- AI model selection for capture filtering, embeddings, and RAG generation.
- Infrastructure and deployment approach.
- Logging and monitoring strategy.

# Glossary

**Sentiora**
The overall permission-based Digital Life Intelligence Platform described in this document.

**Memory**
The module that captures and structures meaningful information from user-approved sources.

**Intelligence**
The module that reasons over Memory via semantic search, RAG, and summarization.

**Footprint**
The (next-phase) module that maps a user's accounts, services, and identity surface.

**Shield**
The (next-phase) module that consumes Footprint and Memory signals to flag risk and recommend protective action.

**Meaningful Capture Engine**
The filtering system that decides whether a page or item is engaging enough (dwell time, scroll depth, revisits) to become a memory.

**Sensitive-content exclusion policy**
A permanent, non-overridable capture-time exclusion covering Incognito mode, authentication/password-sensitive interfaces, payment/financial-sensitive interfaces, and identifiable private health-sensitive content. Exact detection mechanisms are defined in the Technical Specification.

**RAG (Retrieval-Augmented Generation)**
A technique where an AI model retrieves relevant stored content before generating an answer, grounding the response in that content.

**Memory (noun, singular)**
A single structured record capturing one piece of meaningful information, with source, timestamp, and excerpt.

**Grounded answer**
An AI-generated response whose claims are traceable to specific retrieved memories via citations.

# Document Approval / Sign-off

Product requirements in this document are **frozen** as of v2.1 and are ready to be used as the source of truth for design, engineering, and the Technical Specification. This is distinct from formal stakeholder sign-off: approvals below are recorded once each function has actually reviewed the frozen document; names and dates are intentionally left blank pending that review.

| ROLE | NAME | DATE | SIGNATURE |
|---|---|---|---|
| Product Lead | | | |
| Engineering Lead | | | |
| Design Lead | | | |
| AI/ML Lead | | | |